

Traditional Risk Factors versus Novel Cardiometabolic Indices in Predicting High-Risk Plaque and Coronary Calcification in Diabetes Mellitus

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Abstract

Background: Cardiovascular disease is a major complication of Diabetes Mellitus (DM). Computed Tomography Coronary Angiography (CTCA) is widely used to identify High-Risk Plaque (HRP) and coronary artery calcification (CAC). Beyond conventional risk factors, several novel cardiometabolic indices have been proposed, but their real-world clinical value remains debated.

Objective: To compare conventional clinical and laboratory risk factors (age, gender, chronic kidney disease [CKD], hypertension, smoking status, lipid profile, uric acid, HbA1c) against four derivative cardiometabolic indices (TyHbA1c Index, TG/HDL Ratio, Non-HDL Cholesterol, SUA/HDL Ratio) as predictors of HRP and CAC ≥ 100 Agatston.

Methods: A cross-sectional study was conducted on DM patients undergoing CTCA at RSUP Ngoerah, Denpasar, in 2025. All candidate factors were compared bivariate by HRP status, and independent predictors were confirmed using multivariate logistic regression.

Results: A total of 52 DM patients with complete records were analyzed. Sixteen patients (30.7%) were newly diagnosed DM prior to CTCA. Among the 36 known DM cases, 17 (32.7%) were on oral antidiabetic medications, 17 (32.7%) were on insulin, and only 2 (3.8%) received no DM medication. Most patients were referred by cardiologists due to a prior history of cardiac events, and 37 (71.2%) were already on statin therapy. The proportion of HRP on CTCA was found in 71.2% patients. Bivariate analysis showed the HRP-positive group was significantly older (mean age 63.16 vs. 55.8 years, $p < 0.01$) and had a higher calcification burden (median 556.7 vs. 11.6 Agatston, $p < 0.001$) than the HRP-negative group. None of the derivative cardiometabolic indices (TyHbA1c Index, TG/HDL Ratio, Non-HDL Cholesterol, and SUA/HDL Ratio) showed statistically significant differences between the groups ($p > 0.05$). Multivariate analysis confirmed age as the sole independent predictor for both HRP formation (aOR 1.08; $p = 0.045$) and CAC ≥ 100 (aOR 1.10; $p = 0.021$). CKD Stage ≥ 3 indicated a clinical trend of increased HRP risk but did not reach statistical significance.

Conclusion: Age is the primary determinant driving plaque vulnerability and calcification burden in DM patients. The lack of independent predictive value from lipid-based indices may partly reflect indication bias from concurrent statin therapy, warranting caution before adopting these indices as standalone risk-stratification tools.

Keywords:

Cardiometabolic Index; Coronary Artery Calcification; Diabetes Mellitus; High-Risk Plaque.